MARATHON PETROLEUM

Refinery Process Optimization Advisor

A Plain-Language Guide to How the Plant and the AI Agents Work Together

Multi-Agent AI Advisory System

IIM Mumbai · PGPEX Co'27 Capstone Project

Prepared for operators, reviewers, and anyone unfamiliar with refining

Contents

1. Why This Document Exists

This system simulates a real oil refinery and puts a team of AI "advisors" in charge of watching it, 24 hours a day. You do not need any background in chemical engineering to understand this guide — everything is explained from the ground up, with the actual formulas shown so you can see there's no magic involved, just arithmetic.

By the end of this document you should be able to answer three questions: **What does the plant actually do? How do the AI agents decide what to recommend? And how would I, as an operator, actually use this day to day?**

> **In one sentence:** crude oil goes in one end of six connected processing units, and gasoline, diesel, and jet fuel come out the other end — and because the units are all connected, a change in one place always changes something somewhere else. That's the whole reason the AI agents exist: to think through those connections faster and more consistently than a person could, while never overriding hard safety limits.

2. The Big Picture in One Diagram

Here is the entire plant, all six processing units, in the order crude oil actually flows through them:

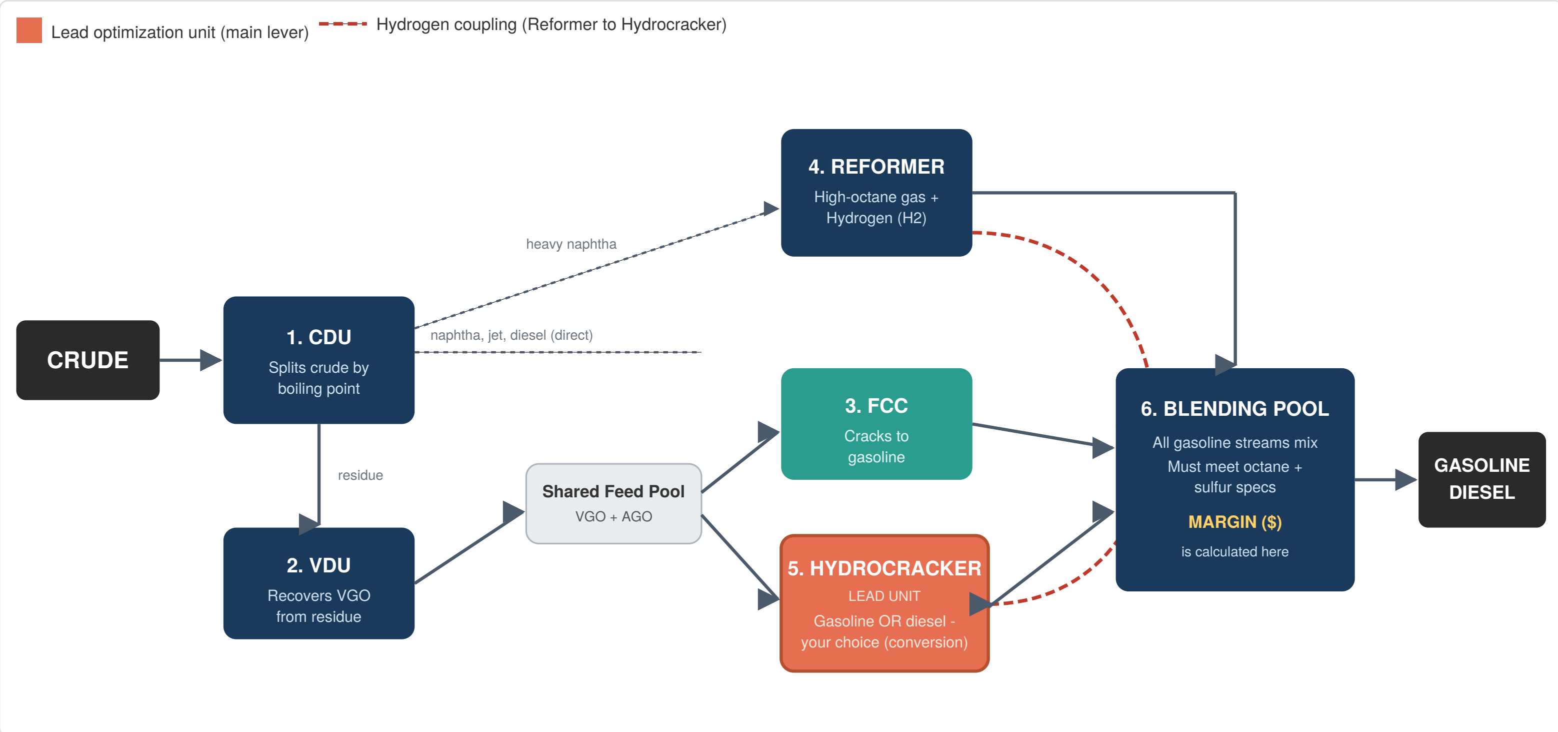

Figure 1 — Crude flows left to right. The orange box (Hydrocracker) is the main lever operators use to choose between making more gasoline or more diesel. The red dashed line shows hydrogen gas flowing from the Reformer to the Hydrocracker — a connection that turns out to matter a great deal (see Section 4).

Two things worth noticing immediately: first, crude splits into several paths right away (Unit 1); second, two of those paths (Units 3 and 5) are drawing from the *same shared pool* of material and are effectively competing with each other. That competition is one of the most important ideas in this whole document.

3. How the Plant Works — Unit by Unit

Each unit below is explained the same way: what it does in plain English, what an operator can control ("knobs"), the actual equation the simulator uses, and the hard safety limit that can never be crossed.

Unit 1 — CDU (Crude Distillation Unit) **The Splitter**

Crude oil is a mixture of many different hydrocarbons. The CDU heats it up and separates it by boiling point — lighter, more valuable material rises off first (naphtha, jet, diesel); the heaviest, least valuable material ("residue") sinks to the bottom.

Knob	Normal Setting	Allowed Range	Effect
Furnace Temperature	365 °C	340–400 °C	Hotter = more diesel, less leftover residue
Throughput	235,000 bbl/day	180,000–250,000	How much crude flows through per day

```
Diesel produced  =  18.5% of crude  +  a small bonus for every degree above 365°C
```

Hard limit: furnace temperature must stay at or below 370°C, or the furnace tubes risk metal damage.

Unit 2 — VDU (Vacuum Distillation Unit)

The CDU's leftover residue still has value trapped in it. The VDU distills it again, but under vacuum — which lets heavy material boil at a lower temperature without burning it — recovering more usable oil ("VGO").

```
VGO recovered  =  40%  +  (32% × how hard the unit is pushed)
```

Hard limit: the "how hard" setting cannot exceed 0.85, or the unit risks forming unwanted coke.

Unit 3 — FCC (Fluid Catalytic Cracker) **Gasoline Engine #1**

The FCC "cracks" large, heavy molecules into smaller gasoline molecules using a catalyst — much like a chemical sledgehammer. It's the traditional way refineries make gasoline.

```
Gasoline produced   =  35%  +  (32% × how hard it's run)
Unwanted coke        =  3%  +  4.5% × (how hard it's run)²    ← grows FAST
```

Notice the coke formula has a "squared" term — push this unit too hard and coke production accelerates quickly. This is deliberately the plant's most sensitive trip-wire, and you'll see it in action in Section 7.

Hard limit: coke production must stay at or below 6.0% of the feed.

Unit 4 — Reformer **Octane + Hydrogen Maker**

The Reformer does two jobs simultaneously: it turns lower-quality naphtha into premium, high-octane gasoline blending material, *and* as a side-effect of that chemistry, it produces hydrogen gas — which the plant needs elsewhere (see Unit 5).

```
Octane quality  =  91  +  (9 × how hard it's run)
Hydrogen made   =  more crude processed × (600 + 500 × how hard it's run)
```

Hard limit: the "how hard" setting cannot exceed 0.88.

Unit 5 — Hydrocracker **THE LEAD UNIT — Main Control Lever**

This is the single most important unit in the plant, and the one the AI agents pay the most attention to. It converts leftover oil into a *mix* of gasoline and diesel — and unlike the other units, the operator directly controls that mix with one setting called **"conversion."**

Conversion Setting	Result
Low (near 0.60)	Mostly diesel
Normal (0.75)	Balanced mix
High (near 0.98)	Mostly gasoline

This one dial is why the Hydrocracker is called the plant's "flexibility engine." Real refineries (including Marathon's own Garyville plant) rely on exactly this unit to shift production toward whichever product is more profitable that week.

```
Hydrogen NEEDED here  =  crude processed × (1200 + 1400 × conversion setting)
This must NEVER exceed the hydrogen the Reformer is making (Unit 4)!
```

Hard limit: conversion setting cannot exceed 0.98, AND hydrogen needed can never exceed hydrogen available — a limit that changes moment to moment depending on what the Reformer is doing.

Unit 6 — Blending Pool

Where Everything Cashes Out

Every gasoline stream from every unit gets mixed together here into one final product, which must meet real quality rules before it can be sold. This is also where the plant's daily profit is calculated.

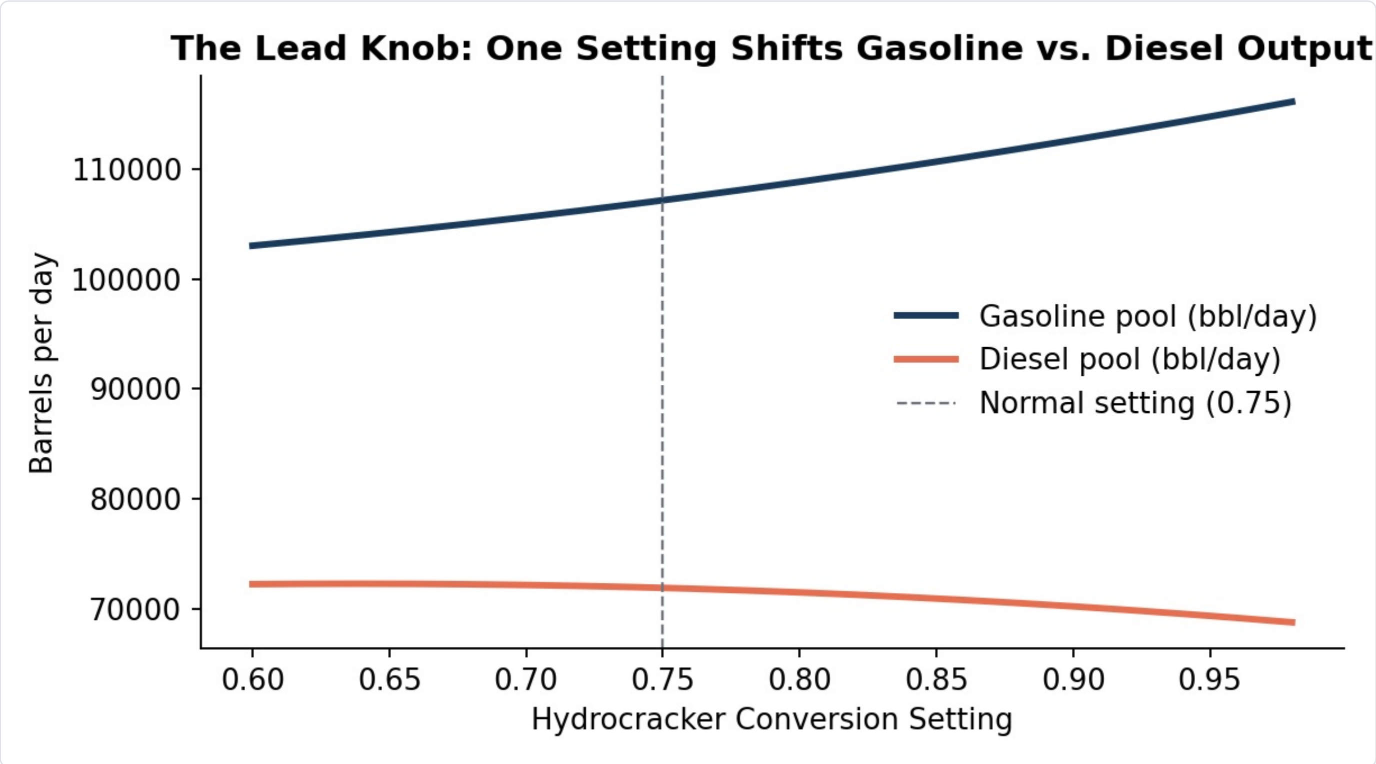

Figure 2 — Moving the Hydrocracker's conversion dial (Unit 5) directly trades gasoline output against diesel output. This single chart is essentially what "optimization" means in this project.

4. Why the Units Can't Be Optimized Alone

If you only look at one unit at a time, you'll make the wrong decision. Here are the three connections that matter most:

Connection A — FCC and Hydrocracker Share the Same Pool of Material

Units 3 and 5 both draw from the same leftover oil. Every barrel sent to the FCC becomes gasoline; the exact same barrel sent to the Hydrocracker could become diesel instead. Which is better depends entirely on which product is selling for more that day — this is the plant's central daily trade-off.

Connection B — The Reformer Feeds Hydrogen to the Hydrocracker

This is the plant's most surprising connection. The Reformer (Unit 4) doesn't just make gasoline — it also produces the hydrogen gas the Hydrocracker (Unit 5) needs to run. Push the Reformer harder for better octane, and you also free up more hydrogen, letting the Hydrocracker run harder too. It's a hidden win-win that a human operator could easily miss, but the AI agents check it every time.

Connection C — Everything Meets Quality Rules at the Blending Pool

You can make an enormous volume of gasoline, but if too much of it came from lower-octane sources, the whole batch can fail to meet the minimum quality standard — making it unsellable. Volume and quality are two different battles, both won or lost at Unit 6.

> **The one-sentence summary of this whole project:** no single unit can be optimized in isolation — a decision at one unit always ripples through to the others, and that ripple effect is exactly what the AI agents are built to reason through.

5. Does Our Simulated Plant Behave Like a Real One?

Before trusting any advice from an AI system, you should ask: is the thing it's advising on actually realistic? We checked this directly — we ran our simulated plant at its normal settings and compared the result against real, published U.S. government data on how actual American refineries perform.

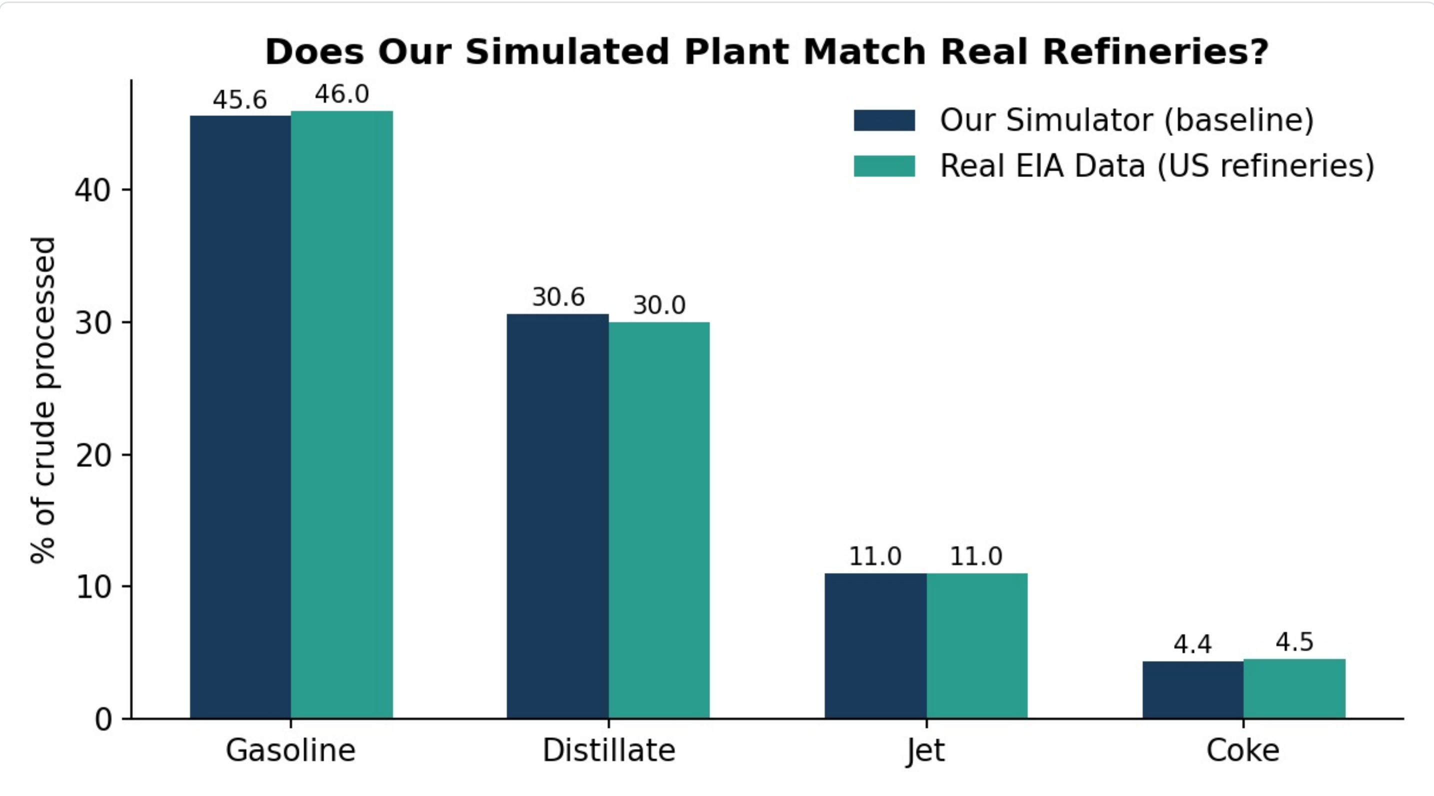

Figure 3 — Our simulator's normal-day output (navy) lands within about half a percentage point of real published U.S. refinery data (teal) on every major product category.

Metric	Our Simulation	Real-World Data	Source
Gasoline produced	45.6%	46.0%	U.S. Energy Information Administration (EIA)
Diesel & heating oil	30.6%	30.0%	
Jet fuel	11.0%	11.0%	
Coke (byproduct)	4.4%	4.5%	
Daily profit margin	**$16.13 / barrel**	**$16.87 / barrel**	**Marathon Petroleum's own 2025 financial filing**

We also generated 180 simulated "operating days," each with slightly different market prices and crude quality, to see whether the plant behaves sensibly across a wide range of conditions rather than just at one lucky setting.

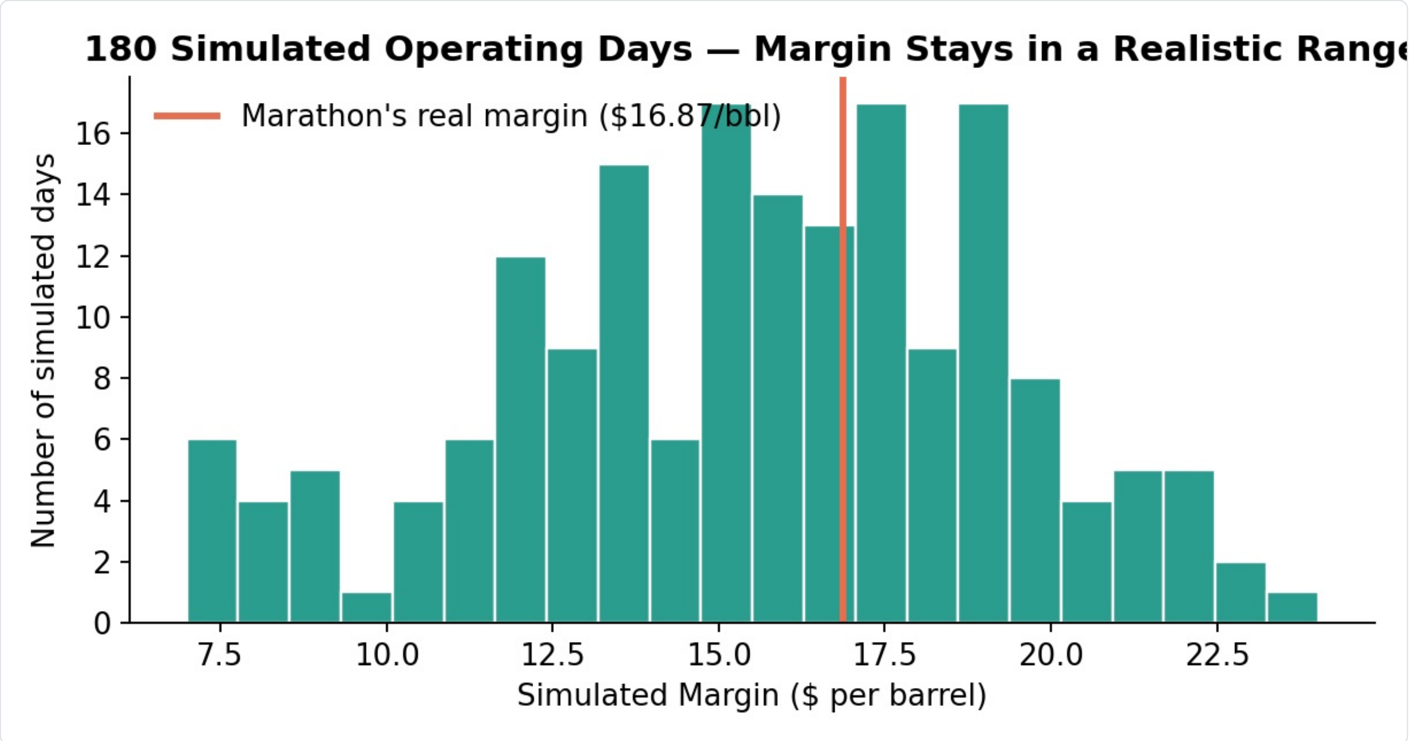

Figure 4 — Across 180 simulated days, profit stays clustered in a believable range around Marathon's real reported margin, with no impossible or broken results.

Bottom line: the simulation is not an arbitrary guess — its default behavior closely matches how real refineries actually perform, and its unit-level formulas are simplified versions of real process engineering relationships (all documented in the companion Operator Manual).

6. Meet the Four AI Agents

Four specialized AI agents watch and advise the plant. Each has exactly one job, and none of them can do each other's job — this separation is deliberate, so it's always clear which agent is responsible for what.

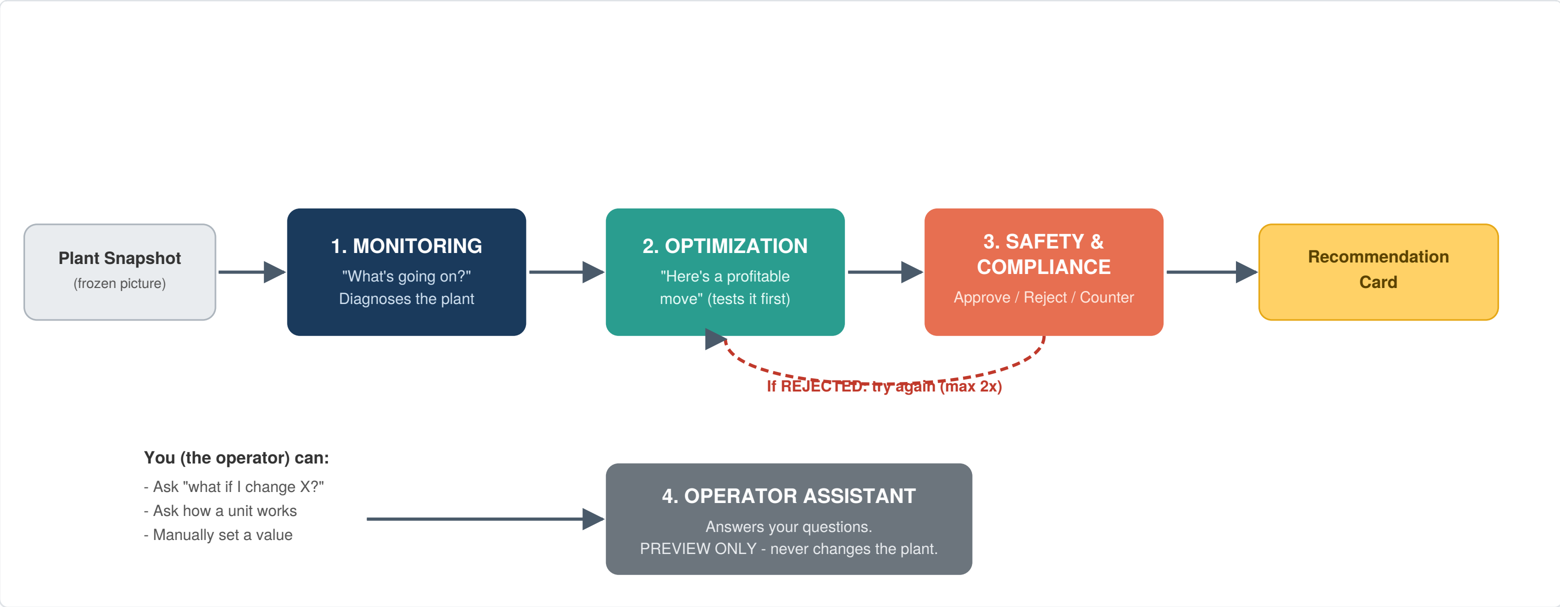

Figure 5 — The advisory pipeline. Notice Safety sits at the end of every chain: nothing reaches the plant without passing its check.

Agent	Job (one sentence)	Can it change the plant?
1. Monitoring	Looks at the current numbers and says what's normal, what's an opportunity, and what's a problem.	No — diagnosis only.
2. Optimization	Proposes a specific, tested change that should increase profit.	No — it proposes; it always tests the idea with the real simulator first.
3. Safety	Checks the proposal against every hard limit before anything happens. Approves, rejects, or asks for a safer version.	**This is the only agent whose decision determines whether a change actually happens.**
4. Operator Assistant	Answers your questions in plain language and previews "what if I changed X?" scenarios.	No — preview only, by design, always.

> **The single most important safety rule in this whole system:** nothing — not an AI proposal, not even a human operator manually typing in a new value — ever reaches the live plant without first passing through the Safety agent's check. There is no back door.

What Makes the Agents' Reasoning Trustworthy

- **They never invent numbers.** Every prediction ("this will raise profit by $4,200/day") comes from actually running the real equations shown in Section 3 — not from the AI's imagination.
- **Every decision cites the actual numbers behind it.** Instead of "this seems risky," Safety says something like: "*furnace temperature would reach 374°C, exceeding the 370°C limit by 4 degrees.*"
- **If the agents can't find a safe improvement, they say so honestly** rather than forcing a risky change — holding current settings is treated as a perfectly good outcome, not a failure.

7. Watching the Agents Work: Two Real Examples

These two examples are taken directly from the system's own test runs — the exact numbers the agents actually produced, not illustrations.

Example A — "There's Money on the Table" (Approve With a Counter-Offer)

Gasoline suddenly becomes more valuable in the market. Here's what happens, step by step:

1. **Monitoring** notices gasoline margins have jumped and the FCC unit (Unit 3) has spare capacity.
2. **Optimization** proposes pushing the FCC hard for more gasoline — predicted gain: **+$228,900/day.**
3. **Safety** checks this against the real formulas and finds a problem: coke production would hit **6.03%, just over the 6.0% hard limit.** It rejects the move and explains exactly why.
4. **Optimization** tries again with a gentler push.
5. **Safety** checks again — this time everything passes. **Approved**, for a smaller but safe gain.

> This is exactly the kind of judgment call a good human operator makes — try for the big win, get told "no, that's too risky," then settle for the safe version instead. The AI does it in seconds, every time, with the same discipline.

Example B — "The Tempting Move Is Actually Unsafe" (Reject, Then Find a Smarter Fix)

A cold snap spikes diesel demand and prices. Here's what happens:

1. **Monitoring** immediately flags a problem: the plant's hydrogen balance is already tight.
2. **Optimization**'s first instinct is the obvious one — raise the furnace temperature and push the Hydrocracker toward diesel. Predicted gain: **+$266,800/day.**
3. **Safety** rejects it — **two** hard limits would be broken at once: furnace temperature *and* the hydrogen balance.
4. **Optimization** comes back with a cleverer idea: instead of forcing the furnace, *raise the Reformer's severity* — which makes more hydrogen available (remember Connection B from Section 4), freeing up room for the Hydrocracker to do its job safely.
5. **Safety approves** the smarter version.

> This example shows the real value of the "network thinking" this whole project is built around — the safe fix didn't come from doing less, it came from understanding a connection between two different units that isn't obvious at first glance.

8. How You, the Operator, Use This System

In practice, there are three simple ways to interact with this system:

1. Just Watch

Leave the plant running. The Monitoring, Optimization, and Safety agents work continuously in the background, checking the plant and recommending small, safe improvements on their own. You don't have to do anything.

2. Ask a Question

Type a question in plain English to the Operator Assistant — things like "*What happens if I raise Hydrocracker conversion to 0.9?*" or "*How does the Reformer affect the Hydrocracker?*" The Assistant answers using the real equations and the manual behind this document — and, importantly, it only shows you a **preview.** Nothing changes until you decide to act.

3. Trigger a Scenario or Set a Value Yourself

You can press a button to simulate a real-world event (a fault, a price spike, a supply problem) and watch the agents react live, or you can manually set a value yourself — and even then, your change is checked by the Safety agent first, exactly like an AI proposal would be. If your manual change is unsafe, it will be blocked and explained — the same protection applies to everyone.

> **Why this design matters:** most "AI advisor" systems either do everything automatically (which operators don't trust) or require an expert to interpret raw numbers (which is slow). This system splits the difference — it explains itself in plain language, shows its work, and never acts without passing through the same safety check every single time, whether the idea came from the AI or from you.

9. Quick Reference Card

All Six Units at a Glance

#	Unit	One-line job	Main knob
1	CDU	Splits crude by boiling point	Furnace temperature
2	VDU	Recovers extra oil from leftovers	Vacuum severity
3	FCC	Cracks oil into gasoline	Cracking severity
4	Reformer	Makes premium gasoline + hydrogen	Reforming severity
5	**Hydrocracker ★**	**Chooses gasoline vs. diesel mix**	**Conversion (THE lead knob)**
6	Blending Pool	Combines everything, checks quality, tallies profit	(no knob — a result)

All Hard Safety Limits

Limit	Maximum Allowed
CDU furnace temperature	370 °C
CDU throughput	250,000 bbl/day
VDU severity	0.85
FCC coke production	6.0%
Reformer severity	0.88
Hydrocracker conversion	0.98
Hydrogen: demand vs. supply	Demand can never exceed supply (recalculated live)
Gasoline quality (octane)	Must be at least 87 (a real U.S. fuel standard)
Diesel sulfur content	Must be at or below 15 parts per million (a real U.S. clean-fuel standard)

The Four Agents at a Glance

Agent	Role
Monitoring	Diagnoses the plant — "what's happening right now?"
Optimization	Proposes and tests profitable changes
Safety & Compliance	The only agent that can approve a change reaching the real plant
Operator Assistant	Answers questions and previews scenarios — never changes anything itself

This document is a companion to the full technical Operator Manual, which contains every equation and configuration value in complete detail for those who want to go deeper.